

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 6 Summer 2025 (Regular)
Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : {FACULTYNAME}
Semester : SEMESTER - 6
Subject Code & Name: BTECPC601 - INTERNET OF THINGS

Time : 3-Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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Which of the following is *not* a characteristic of IoT?

- A) Scalability
- B) Interoperability
- C) Offline Operation
- D) Heterogeneity

REST API is based on which protocol?

- A) FTP
- B) HTTP
- C) SMTP
- D) TCP/IP only

3 Which technology helps IoT handle and analyze large amounts of data?

- A) Bluetooth
- B) Cloud computing
- C) Big data analytics
- D) Embedded systems

4 Which data type in C is used to store a character?

- A) int
- B) char
- C) float
- D) string

5 What is the function of a “for” loop in C?

- A) Making decisions
- B) Calling functions
- C) Repeating tasks
- D) Declaring variables

6 The NodeMCU board is based on which chip?

- A) ATmega328
- B) ESP32
- C) ESP8266
- D) STM32

7. **NodeMCU supports programming in which IDE with Arduino core?**
 A) PyCharm
 B) Eclipse
 C) Arduino IDE
 D) Keil
8. **Which data type is used in Python to store a group of items in key-value pairs?**
 A) List
 B) Tuple
 C) Set
 D) Dictionary
9. **Which command is used in Linux to check internet connectivity?**
 A) cd
 B) dir
 C) ping
 D) copy
10. **MQTT stands for:**
 A) Message Query Transfer Technology
 B) Message Queue Telemetry Transport
 C) Multi Query Transfer Tool
 D) Multiple Quality Transfer Technology
11. **Which protocol is specifically designed for low-energy devices like wearables?**
 A) Ethernet
 B) BLE
 C) UART
 D) AMQP
12. **What is the role of GPIO pins in Raspberry Pi?**
 A) Power management
 B) RAM control
 C) Input/Output operations
 D) Booting the OS
- Q2. Solve the following.
- A) Explain the characteristics of IoT? 6
- B) Compare Python Lists and Dictionaries with examples? 6
- Q3. Solve the following.
- A) Explain the steps to interface NodeMCU with a web service? 6
- B) Compare MQTT and HTTP in context of IoT communication? 6
- Q4. Solve Any Two of the following.
- A) **Explain the logical design of IoT?** 6
- B) **Explain enabling technologies for IoT?** 6

- C) **Write a C program to find the sum of the first 10 natural numbers using a loop?** 6
- Q5. Solve Any Two of the following.
- A) **Write a Python program to find the largest of three numbers?** 6
- B) **Explain GPIO and how to interface sensors with Raspberry Pi?** 6
- C) **Explain how to publish data to a web server using NodeMCU?** 6
- Q6. Solve Any Two of the following.
- A) **Explain MQTT protocol with diagram and use case?** 6
- B) **Compare BLE, UART, and Ethernet in IoT applications?** 6
- C) **Explain steps to configure Wi-Fi in Raspberry Pi?** 6